

Business Insights

Customer Segmentation Analysis

Project: Online Retail Customer Segmentation using RFM Analysis

Executive Summary

This project analyzed customer purchasing behavior using the RFM (Recency, Frequency, Monetary) model to identify high-value customers, understand buying patterns, and support data-driven marketing decisions. Data was cleaned in Excel, transformed and analysed in SQL Server, and visualized in Power BI through an interactive dashboard.

Key Findings

1. High-Value Customers Generate Most Revenue

Customers classified as **Champions** and **Loyal Customers** contributed the largest share of total revenue. These customers purchase frequently, spend significantly more than other groups, and have made recent purchases.

Business Recommendation

- Offer exclusive rewards and VIP loyalty programs.
 - Provide early access to promotions and new products.
 - Maintain personalized communication to encourage repeat purchases.
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2. Lost Customers Present a Re-engagement Opportunity

Customers in the **Lost Customers** segment have not purchased for a long period and generally show low purchase frequency.

Business Recommendation

- Launch win-back campaigns.
 - Offer limited-time discounts.
 - Send personalized reminder emails based on previous purchases.
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3. Potential Loyalists Can Become High-Value Customers

Potential Loyalists have purchased recently but have not yet developed consistent purchasing behavior.

Business Recommendation

- Offer personalized product recommendations.
 - Introduce loyalty points after additional purchases.
 - Send targeted promotions to encourage repeat buying.
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4. Customer Spending Varies Across Segments

Average customer spending differs considerably between segments. Champions consistently show the highest monetary value, while At-Risk and Lost Customers contribute much less revenue.

Business Recommendation

- Allocate marketing resources toward high-return customer segments.
 - Reduce spending on low-performing campaigns.
 - Develop personalised retention strategies.
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5. Customer Retention Is More Cost-Effective Than Acquisition

The analysis shows that increasing retention among existing loyal customers is likely to generate greater returns than focusing solely on acquiring new customers.

Business Recommendation

- Strengthen customer engagement initiatives.
 - Monitor changes in Recency to identify customers at risk of churn.
 - Track customer lifetime value regularly.
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Dashboard KPIs

The dashboard provides decision-makers with the following metrics:

- Total Revenue
- Total Customers

- Total Orders
 - Average Order Value
 - Average Recency
 - Average Frequency
 - Average Monetary Value
 - Revenue by Customer Segment
 - Customer Distribution by Segment
 - Monthly Sales Trend
 - Top Customers
 - Geographical Sales Distribution
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Business Value

The project enables organizations to:

- Identify profitable customers.
 - Improve customer retention.
 - Optimize marketing campaigns.
 - Increase customer lifetime value.
 - Support evidence-based decision-making.
 - Monitor customer behavior through interactive dashboards.
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Conclusion

Using RFM segmentation provides a clear understanding of customer purchasing behavior. The insights generated from this analysis can help businesses increase revenue, improve customer retention, and develop targeted marketing strategies based on customer value.